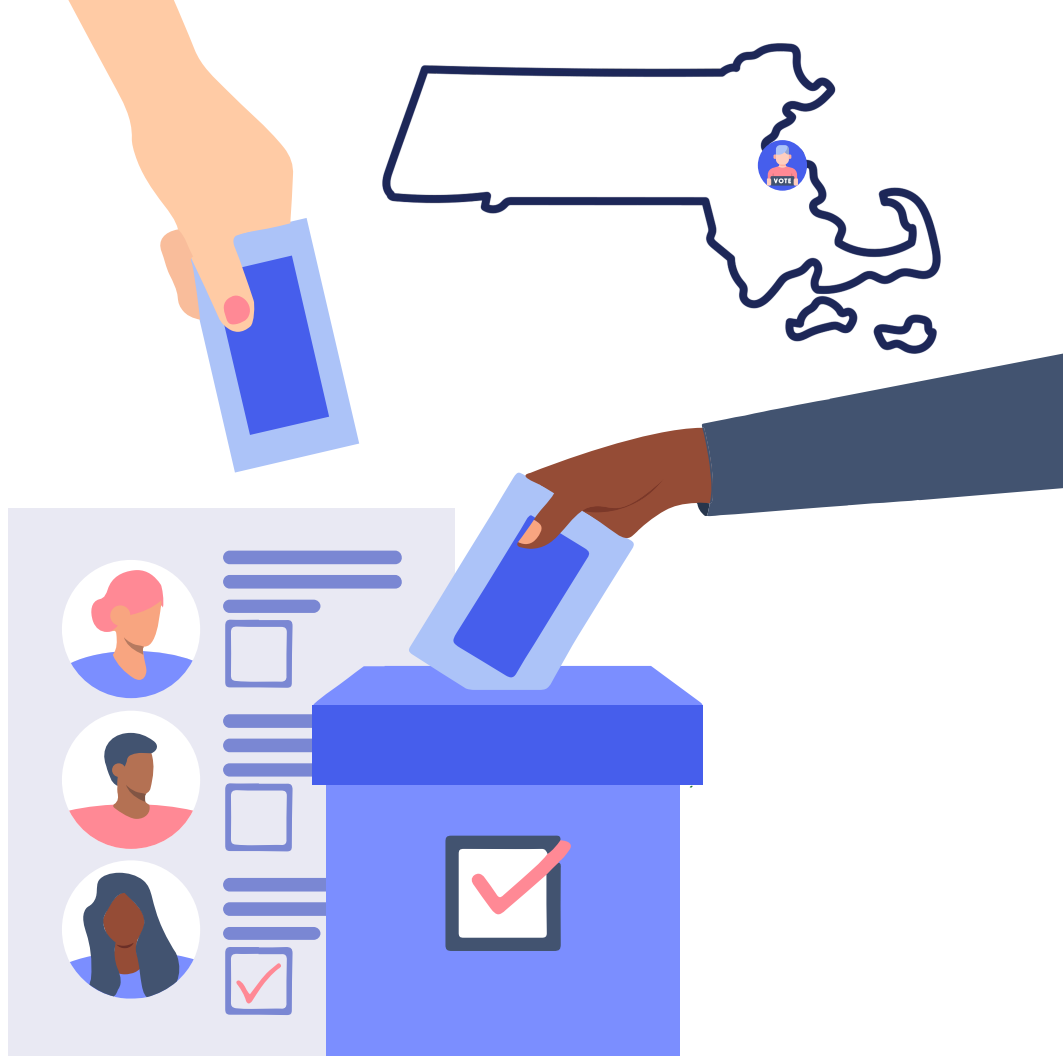


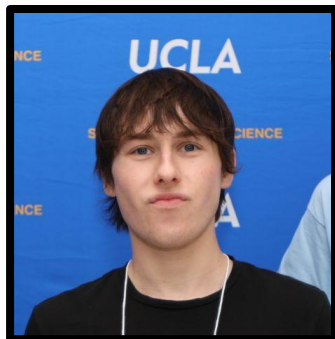
# Asymmetric Voter Turnout Across Elections in Quincy, MA

## *Wil-Cox Transformations*

Lawrence Dzhandzhugazov, Phoebe Chen, Samuel Reade, Sooyeon Byun, and Montgomery Soladay



# Wil-Cox Transformations



**Lawrence Dzhandzhugazov**  
Statistics & Data Science  
Central & East European Studies



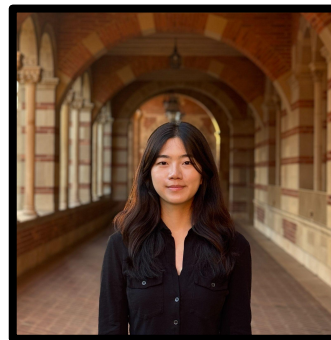
**Phoebe Chen**  
Statistics & Data Science



**Samuel Reade**  
Statistics & Data Science



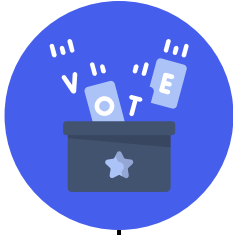
**Montgomery Soladay**  
Statistics & Data Science



**Sooyeon Byun**  
Statistics & Data Science

# The Problem

**Low Local Election Turnout** weakens **democracy**



1

## Local issues decided by few

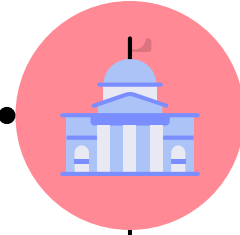
Low election turnout → local measures passed without the people most affected voting.



2

## Cherry-Picking

Candidates can target or represent certain segments, if they know they will vote for them.



3

## Lack of Representation

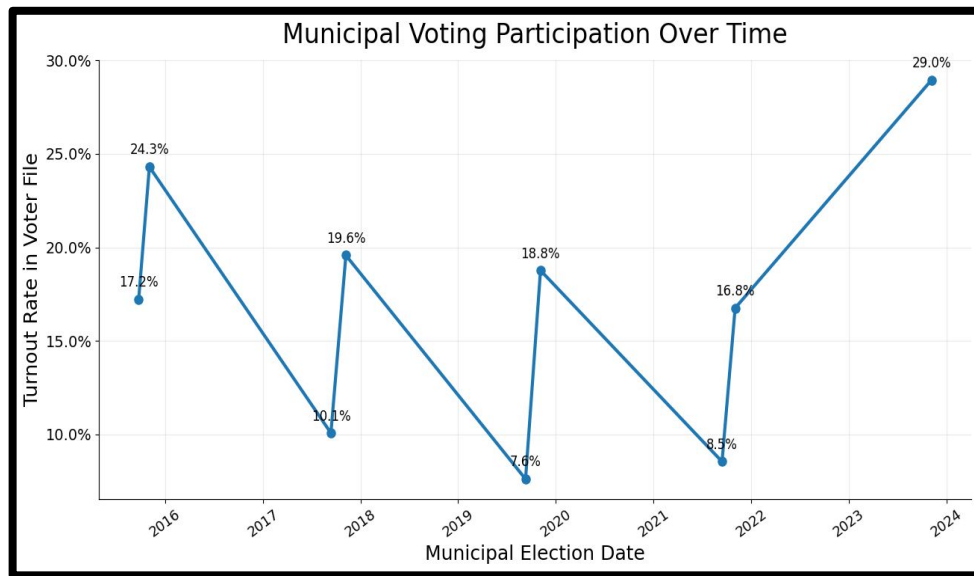
Unpopular legislation passes, even if they are opposed by the majority.

# Recent Elections

## Quincy Voter and Election Context

- Quincy conducts **municipal, state, and federal elections** across **31 precincts**.
- **Federal and presidential elections** generally attract much higher participation than municipal elections.
- **~30%** of registered voters participated in Quincy's 2023 city election.
- Turnout also varies substantially across wards and precincts, meaning some neighborhoods are **more represented** in local decisions than others.
- Prior QuincyVotes research suggests turnout differences are associated with factors such as **homeownership, neighborhood demographics, age, and residential stability**.

Sources: Quincy Election Department; QuincyVotes, 2023 Voting Data; QuincyVotes, Reducing Disparities in Voter Turnout.

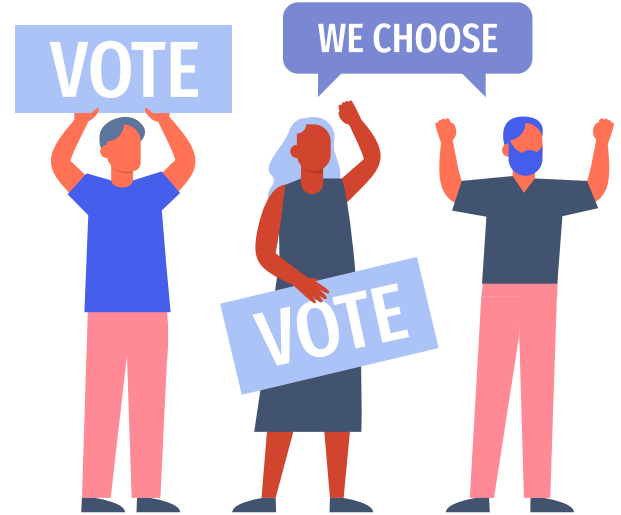


# Research Question

Our analysis therefore focuses on identifying:

- voters who participate **federally** but not **municipally**;
- precincts with unusually **high or low local turnout**; and
- demographic, geographic, and voting-history factors associated with municipal participation.

**How can Quincy better incentivize voter turnout for the 2027 mayoral election, given evidence of higher local turnout in previous elections?**



# Dataset

## Data Cleaning & Missingness

- **33,315 rows**, representing registered voters in the city of Quincy
- Started with **900 variables** describing voter demographic information from the L2 voter file
- **Removed 294 completely empty columns** and variables, as well as columns with >80% missingness, personally identifiable information (PII), **multicollinearity**, and **leakage**
- **Consumer behavior variables** showed the highest missingness
- Eventually also removed columns that did not contribute to models

**Top Variables by Missingness in the Cleaned Voter File**

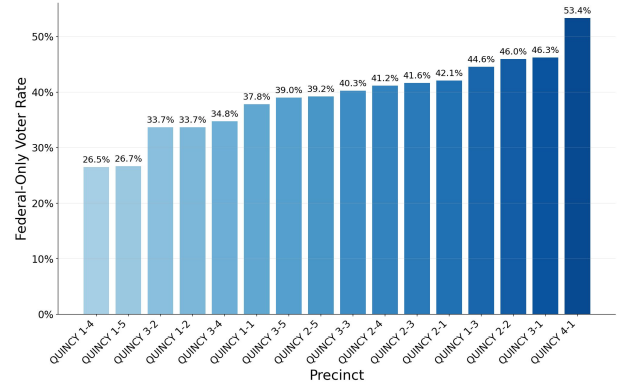
| Variable                                                | Missing Percent |
|---------------------------------------------------------|-----------------|
| ConsumerDataLL Palin Sarah                              | 100.0%          |
| ConsumerDataLL Bachmann Michelle                        | 100.0%          |
| ConsumerDataLL Cain Herman                              | 100.0%          |
| ConsumerDataLL McCain John                              | 100.0%          |
| ConsumerDataLL Church Attendee                          | 100.0%          |
| ConsumerDataLL Bush George                              | 100.0%          |
| ConsumerData African American Professional in Household | 100.0%          |
| ConsumerDataLL Clinton Hillary                          | 100.0%          |
| ConsumerData Donor Environmental Issues                 | 100.0%          |
| ConsumerDataLL Election of Conservative Judges          | 100.0%          |
| ConsumerData High End Appliances                        | 100.0%          |
| ConsumerDataLL Romney Mitt                              | 100.0%          |
| ConsumerDataLL Gun Control                              | 99.9%           |
| ConsumerDataLL Obama Barack                             | 99.9%           |
| Mailing Addresses NCOA CancelDate                       | 99.9%           |

# Exploratory Data Analysis

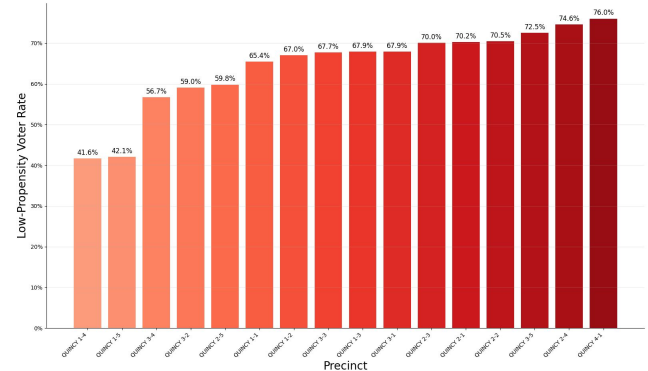
## Geographic Turnout Patterns

- Municipal turnout challenges are not evenly distributed across Quincy
- Quincy 4-1 had the **highest concentration** of low-propensity municipal voters(76.0%) and federal-only voters (53.4%)
- Federal-only voters : these voters are politically engaged but locally inactive  
→ a **key opportunity** for targeted outreach
- **Precinct-level outreach** may be an effective strategy for increasing participation in future municipal elections

Federal-Only Voter Rate by Precinct



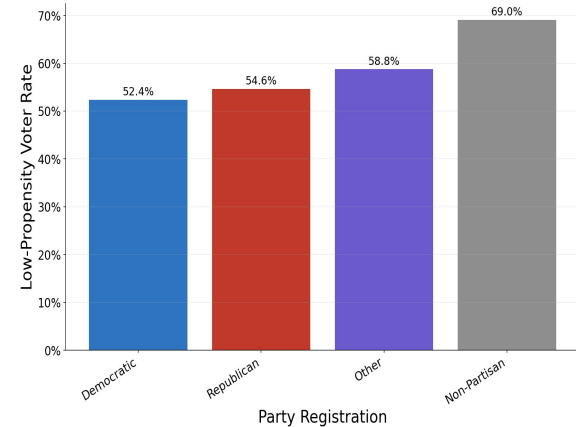
Low-Propensity Municipal Voter Rate by Precinct



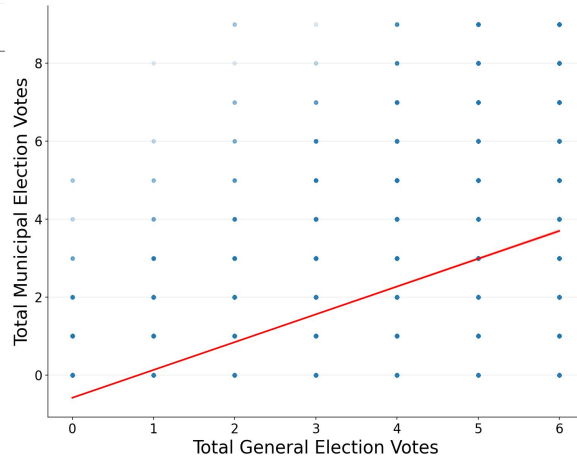
# Exploratory Data Analysis

## Low-Propensity by Party Registration and Voting History

Low-Propensity Municipal Voter Rate by Party Registration

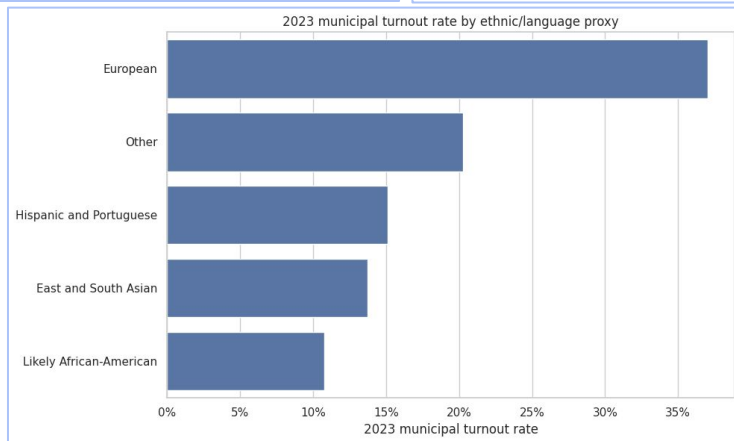
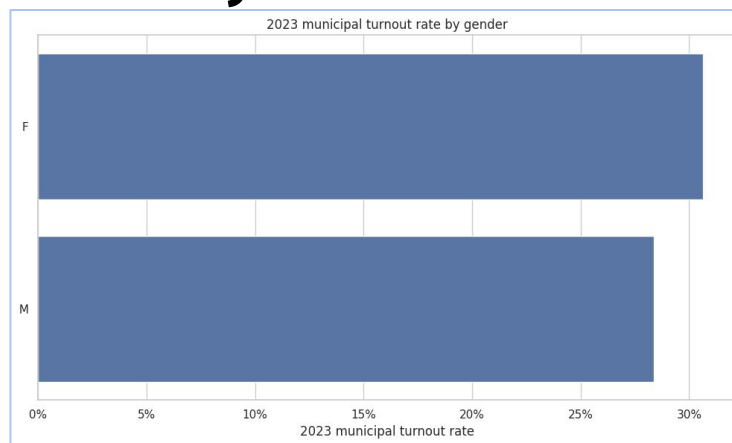
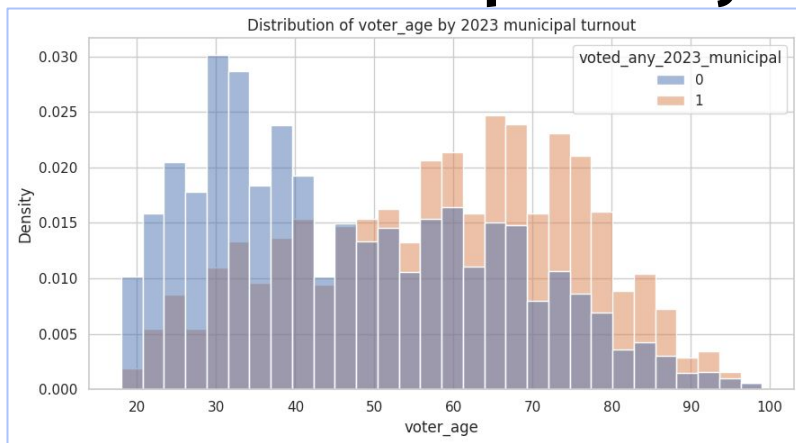


General Election Voting vs. Municipal Election Voting



- Non-partisan voters have the highest low-propensity rate (69.0%)
- Democratic (52.4%) and Republican (54.6%) voters showed higher municipal participation
- Non-partisan voters may be a key outreach target for increasing municipal election participation
- People that vote in general elections, generally vote in municipal as well.

# Exploratory Data Analysis

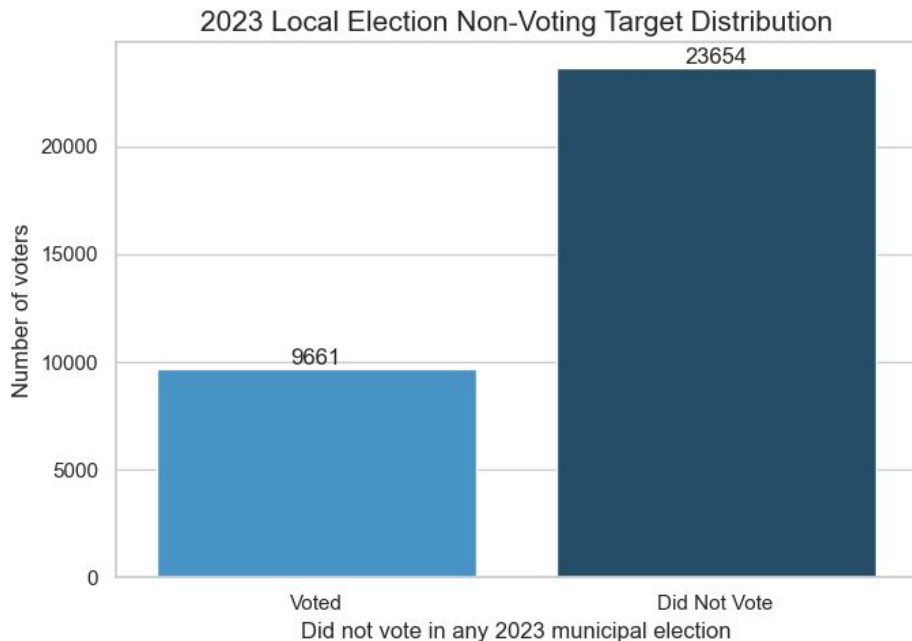


★ Additional voter/non-voter **demographic discrepancies** (based on the 2023 municipal elections turnouts)

# Methodology

**Goal:** predict municipal non-voting risk in Quincy.

**Target variable:** **DidNotVote** = 1 if the voter did **not** vote in any 2023 Quincy municipal election.



- **DidNotVote** = 0 if the voter voted in **at least one** 2023 municipal election

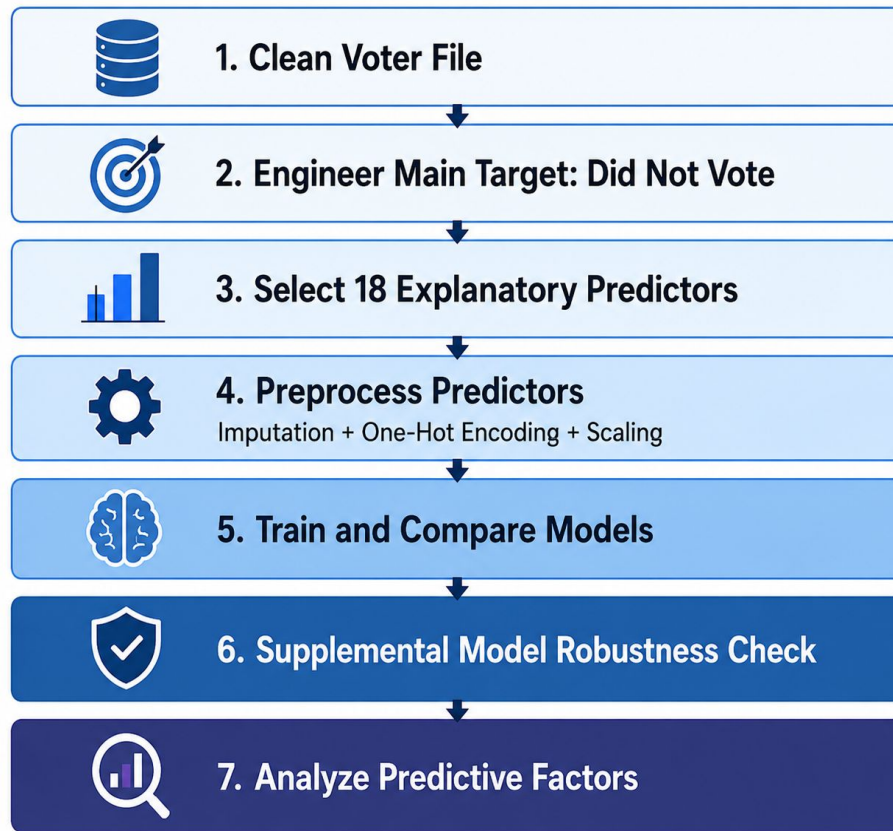
- Predicted probabilities were converted into **four outreach risk groups**

- Direct vote-history and voter-frequency predictors were **excluded** to reduce leakage

- Final model focused on explanatory demographic, socioeconomic, political, residential, and geographic predictors

# Methodology

- Final reduced model used 18 explanatory predictors removing 15 from the original model
- ZIP, precinct, and ward were treated as categorical variables; categorical predictors were one-hot encoded and numeric predictors were scaled/imputed
- Models were trained with sklearn pipelines using a held-out test set
- **A supplemental model used an alternative age/registration adjusted low-propensity municipal turnout target as a robustness check**
- Analyzing Predictors for Importance was the final step



# Model Selection

- We also tested a larger model containing **prior voting history** and voting eligibility-adjusted **turnout measurements**.
- It predicted turnout more accurately, but voting history dominated the results.
- Its main finding was that voting is habitual.
- We selected the reduced models for **clearer, more actionable interpretation**.

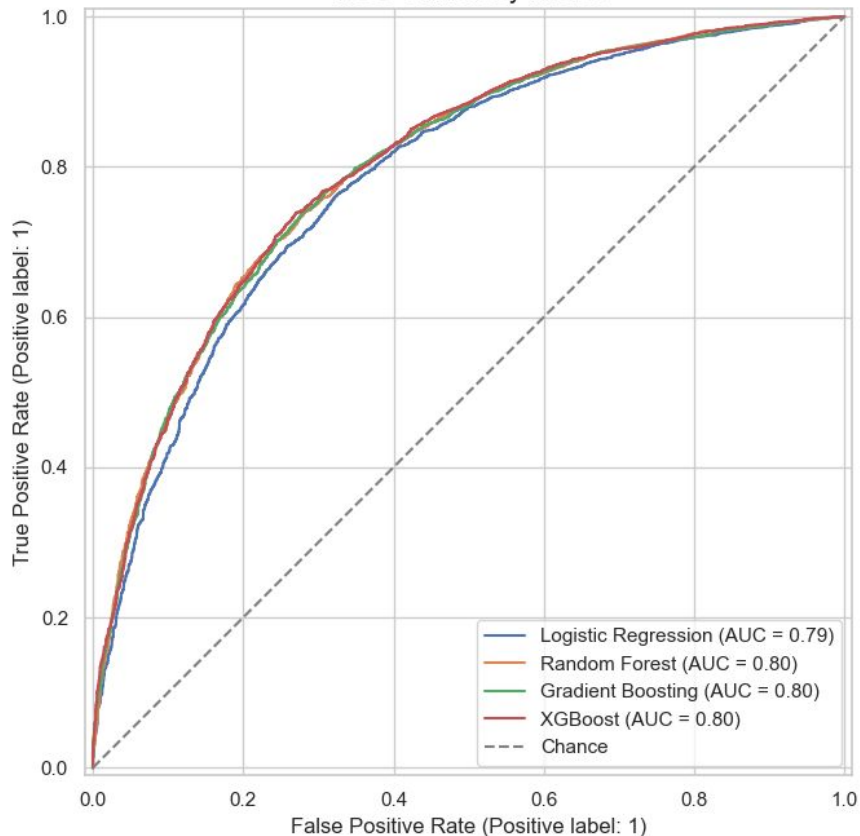
Larger Model: Top Variable Importances (XGBoost)

| Rank | Predictor                                          | Importance |
|------|----------------------------------------------------|------------|
| 1    | Any prior municipal vote                           | 32.7%      |
| 2    | Prior overall turnout rate                         | 9.8%       |
| 3    | Eligibility-adjusted municipal turnout rate        | 4.2%       |
| 4    | Eligibility-adjusted general-election turnout rate | 1.9%       |
| 5    | Eligibility-adjusted overall turnout rate          | 1.6%       |
| 6    | Modeled homeowner status                           | 1.1%       |

The top five variables were all based on prior voting behavior and together accounted for approximately 50.3% of the model's total importance.

# Model Evaluation

ROC Curves by Model



| Model               | Accuracy | Precision | Recall | F1    | AUC   |
|---------------------|----------|-----------|--------|-------|-------|
| XGBoost             | 0.777    | 0.808     | 0.899  | 0.851 | 0.803 |
| Random Forest       | 0.741    | 0.857     | 0.763  | 0.807 | 0.801 |
| Gradient Boosting   | 0.774    | 0.805     | 0.900  | 0.850 | 0.800 |
| Logistic Regression | 0.720    | 0.856     | 0.728  | 0.787 | 0.786 |
| Supplemental Model  | 0.728    | 0.741     | 0.897  | 0.812 | 0.729 |

- Compared **four supervised classifiers** using a held-out test set
- Reduced **XGBoost** had the strongest ROC-AUC among reduced models
- Recall was high, meaning the model identified many actual non-voters
- The supplemental model showed similar recall, supporting the broader non-voting risk framework

# Model Evaluation

**Overall Accuracy:** **77.7%** of voters were correctly classified

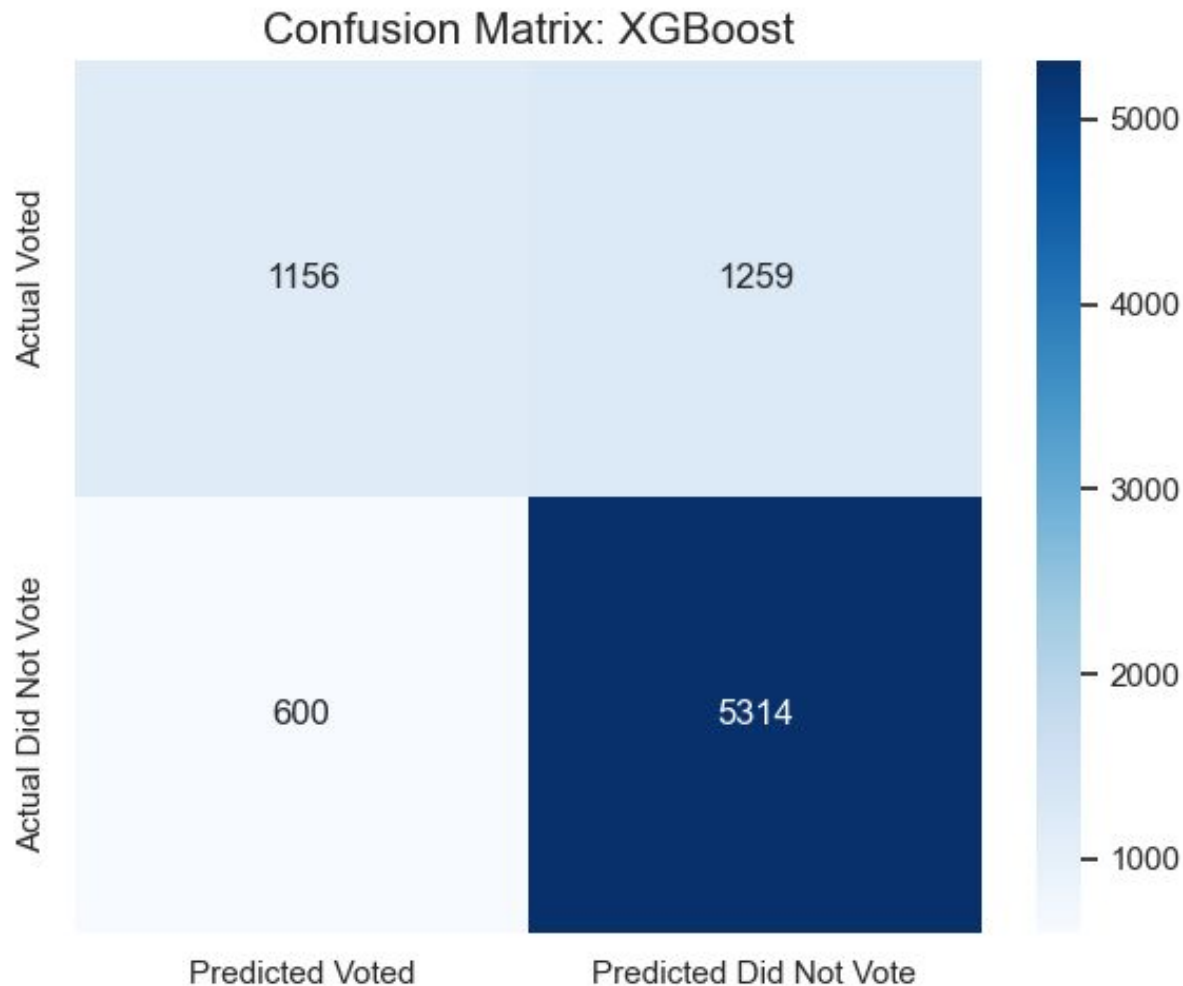
**ROC AUC:** Approximately **0.80**, indicating strong ability to distinguish voters from non-voters

**Non-Voter Recall:** **89.9%** of actual non-voters were correctly identified

**Non-Voter Precision:** **80.8%** of individuals predicted to be non-voters were truly non-voters.

**F1 Score (Non-Voters):**

Approximately **0.85**, demonstrating a strong balance between precision and recall.



# Important Findings

| Risk Group                | Voters | Actual | Predicted |
|---------------------------|--------|--------|-----------|
| Low Non-Voting Risk       | 1,364  | 0.144  | 0.204     |
| Moderate Non-Voting Risk  | 5,769  | 0.364  | 0.370     |
| High Non-Voting Risk      | 7,525  | 0.630  | 0.640     |
| Very High Non-Voting Risk | 18,657 | 0.891  | 0.881     |

Actual non-voting increased steadily across the predicted risk groups.

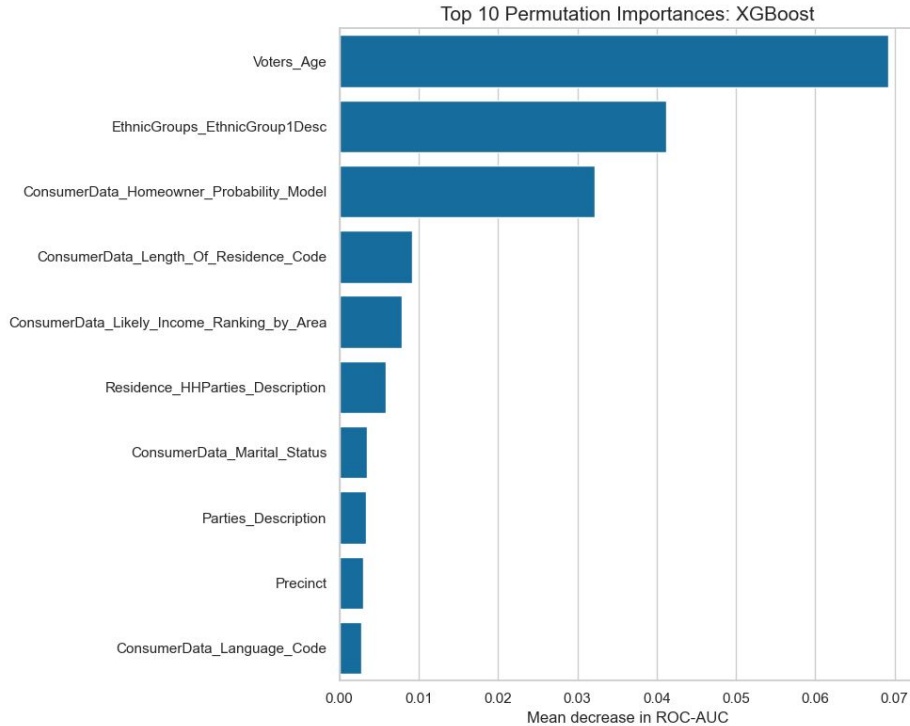
**Low-risk** voters had only **14.4%** actual non-voting.

**Very high-risk** voters had **89.1%** actual non-voting.

The model is most useful as a ranking tool for outreach prioritization.

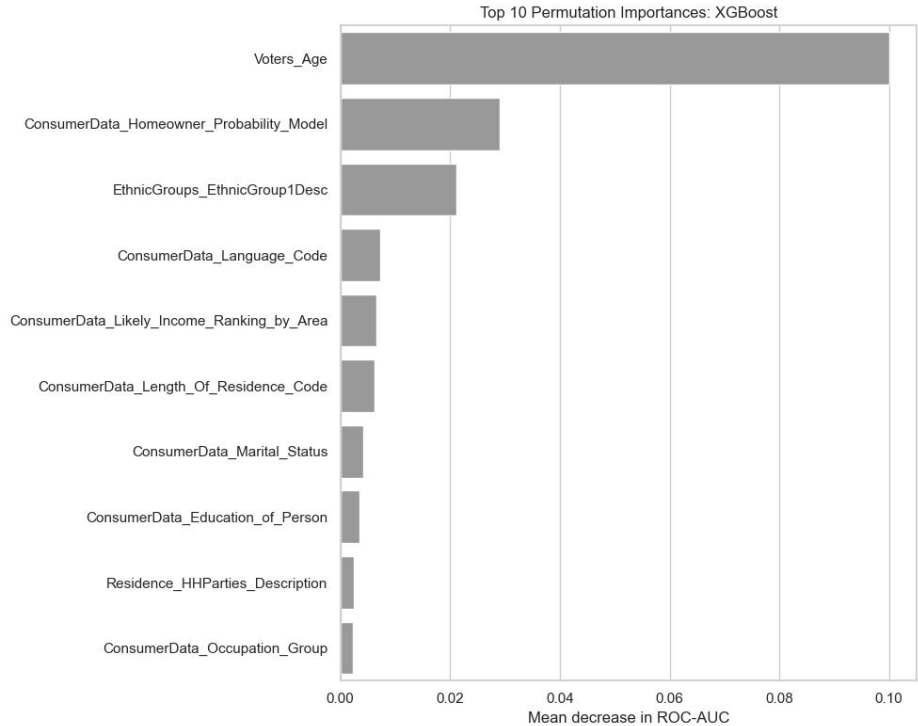
**High** and **very high-risk** voters represent the clearest targets for turnout intervention.

# Important Findings



- **Age** was the strongest predictor in both models

- **Ethnicity and homeownership** were consistently among the most influential factors

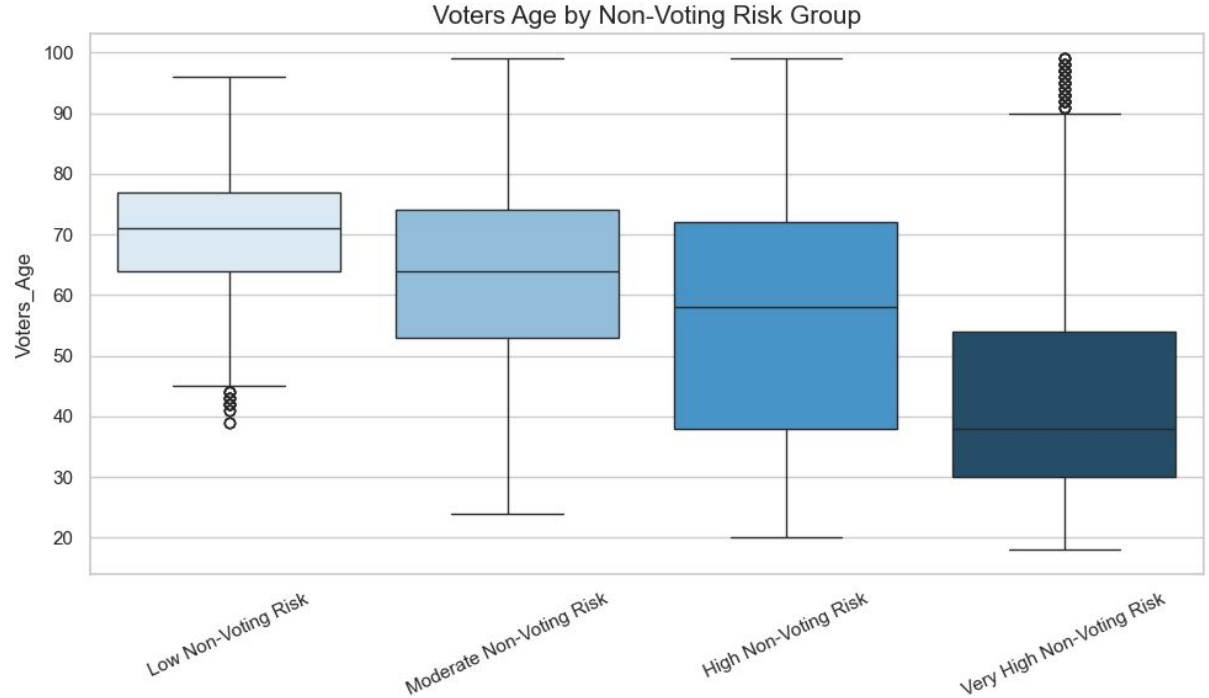


- Supplemental model showed similar predictors

- Model consistency supports that municipal non-voting risk is driven by these factors

# Important Findings

- **Younger** voters were most concentrated in high-risk categories
- **18-24: 99.7%** high or very high risk
- **25-34: 98.7%** high or very high risk
- **35-44: 89.8%** high or very high risk
- **Non-partisan** voters made up **68.5%** of the high-risk population
- **Renters** had **95.8%** high-risk rate vs. **47.6%** for **homeowners**
- **Ward 2** and **Ward 4** showed elevated non-voting risk



# Conclusion

- The reduced XGBoost model turned voter patterns into a **practical ranking**
- Predicted risk groups **closely matched** actual 2023 non-voting behavior
- The model is useful for **prioritizing outreach**, but not for proving causation or perfectly predicting individual behavior in future elections — could be improved given more data



# Recommended Next Steps

- Broader issue of **incentivizing voting overall**
- Prioritize voters and precincts where municipal **non-participation risk is highest**
- **Test outreach** through precinct-based, school, workplace, and community spaces



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**Thank you!**